A BOUNDED FOCK-SPACE TOEPLITZ OPERATOR WITH UNBOUNDED BEREZIN–HEAT TRANSFORM AT $t = 1/4$

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Abstract. We record Looi's 2022 construction of a measurable symbol f on $\mathbb{C}$ whose Toeplitz operator T_f is bounded on the Bargmann–Fock space $F^2(\mathbb{C})$, while its heat transform $\widetilde{f}_t$ fails to be bounded at $t = 1/4$.

1. Main Refutation

Theorem 1.1 (Looi, 2022)**.** *There exists a radial/oscillatory measurable symbol $f : \mathbb{C} \to \mathbb{C}$ such that the Toeplitz operator T_f extends to a bounded linear operator on the Fock space $F^2(\mathbb{C})$, whereas the heat transform at time $t = 1/4$ is unbounded on $\mathbb{C}$.*

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